Concepts of model-based quantitative decision making

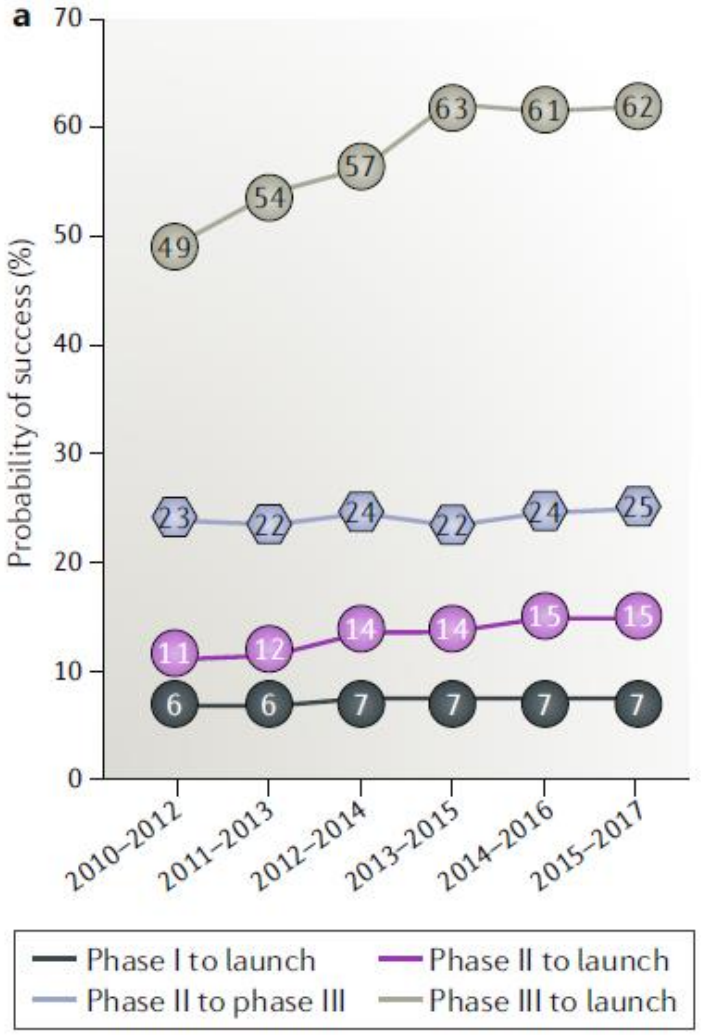
ECP 8506 Clinical Trial Simulation
Sep 18, 2026

Ya-Feng (Jay) Wen, PharmD, PhD
jay.wen3@gilead.com | wenxx164@umn.edu
Associate Clinical Pharmacology Director, Gilead Sciences

Learning Objectives

- Review drug development processes and reasons for drug attrition
- Understand different quantitative medicine approaches
- Utilize credibility assessment framework in model-informed drug development (MIDD)
- Examples of MIDD

About 90% of clinical drug development fails



Dowden H, Munro J. Trends in clinical success rates and therapeutic focus. *Nat Rev Drug Discov*. 2019;18(7):495-496

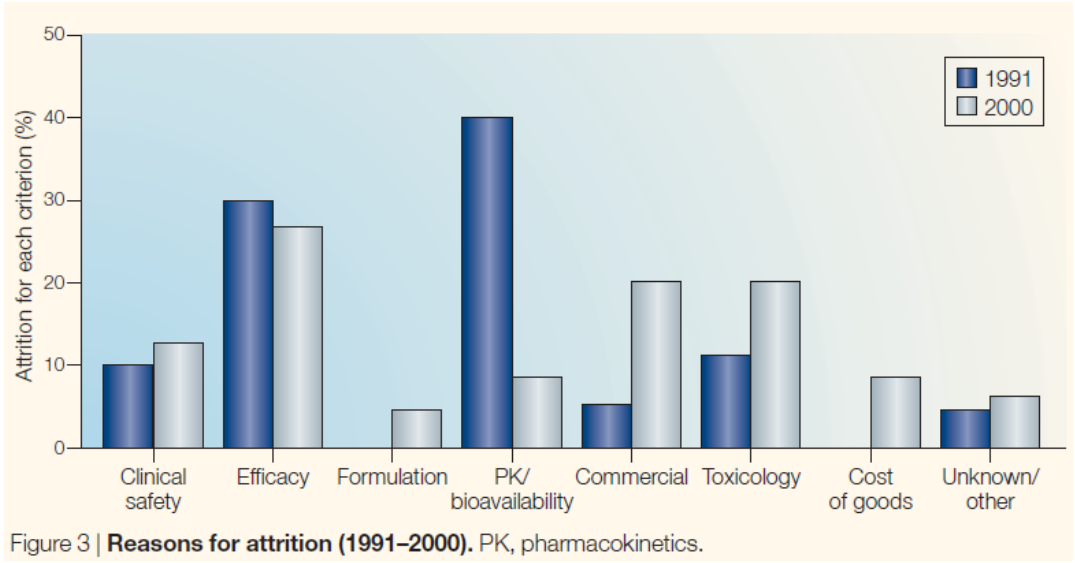
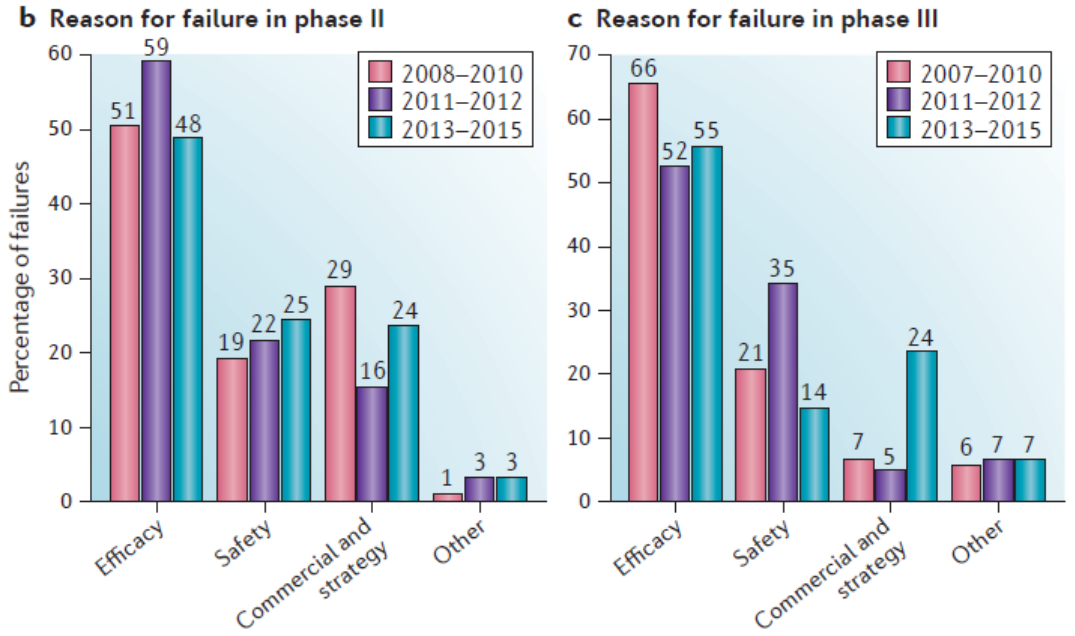


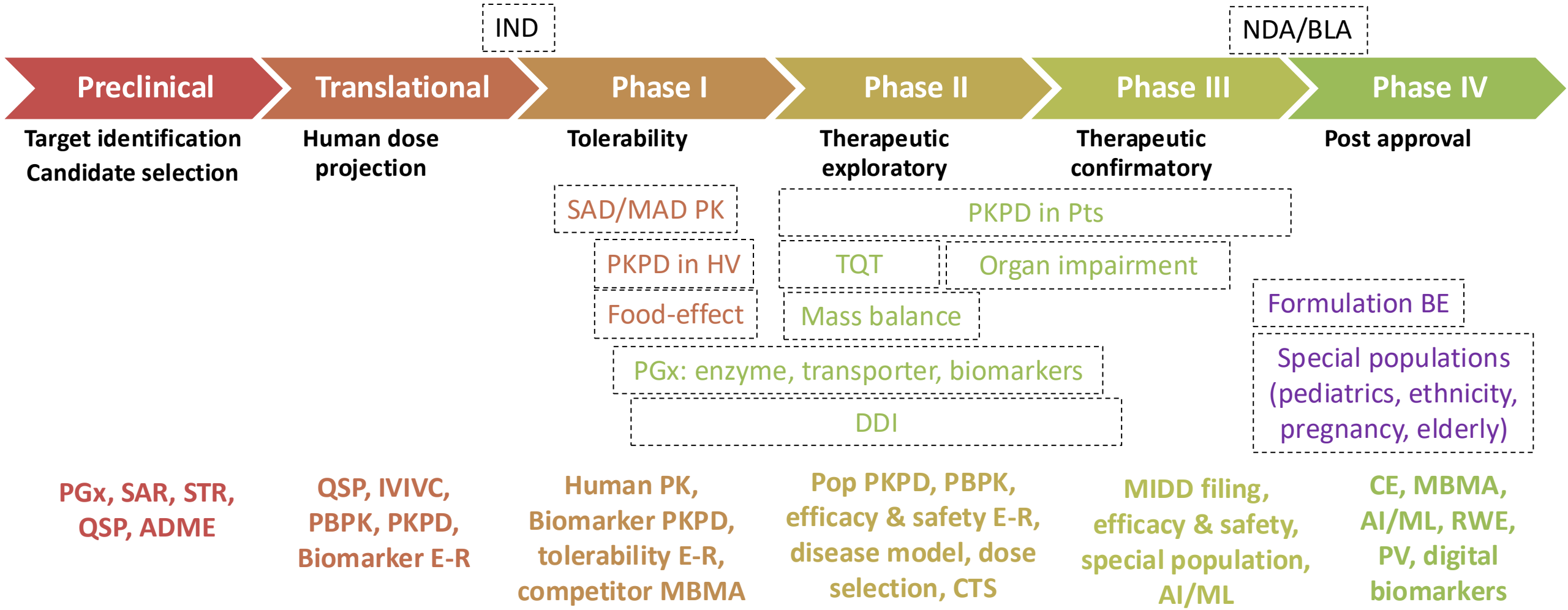
Figure 3 | **Reasons for attrition (1991–2000).** PK, pharmacokinetics.

Kola I, Landis J. Can the pharmaceutical industry reduce attrition rates?. *Nat Rev Drug Discov*. 2004;3(8):711-715.



Harrison RK. Phase II and phase III failures: 2013-2015. *Nat Rev Drug Discov*. 2016;15(12):817-818.

Various tools can be used to support drug development



CE, comparative effectiveness; CTS, clinical trial simulation; E-R, exposure-response; IVIVC, in vitro-in vivo correlation; MBMA, model-based meta-analysis; PBPK, Physiologically based pharmacokinetic; PKPD, pharmacokinetic—pharmacodynamic; PV, pharmacovigilance; QSP, quantitative system pharmacology; RWE, real-world evidence; SAR, structure–activity relationship; STR, structure–tissue exposure/selectivity relationship

Adapted from Dr. Mark Gastonguay. Model-Informed Decision Making in Drug Development. C-Path MIDD Training Course.

Using Quantitative Medicine approach has the potential to maximize patient and societal benefit

- On 03/25/2024, FDA's Center for Drug Evaluation and Research (CDER) launched the new CDER Quantitative Medicine (QM) Center of Excellence (CoE)
  - The development and application of **exposure-based**, **biological**, and **quantitative modeling and simulation** approaches derived from nonclinical, clinical, and real-world sources to inform drug development, regulatory decision-making, and patient care
  - Model-informed drug development (MIDD)
  - Complex innovative trial design (CID)
  - Fit-for-purpose initiative (FFP)
  - Model integrated evidence (MIE)
  - Physiology based biopharmaceutics modeling (PBBM)

https://www.fda.gov/about-fda/center-drug-evaluation-and-research-cder/cder-quantitative-medicine-center-excellence-qm-coe

Examples of fit-for-purpose initiative

University of Minnesota
Driven to Discover®

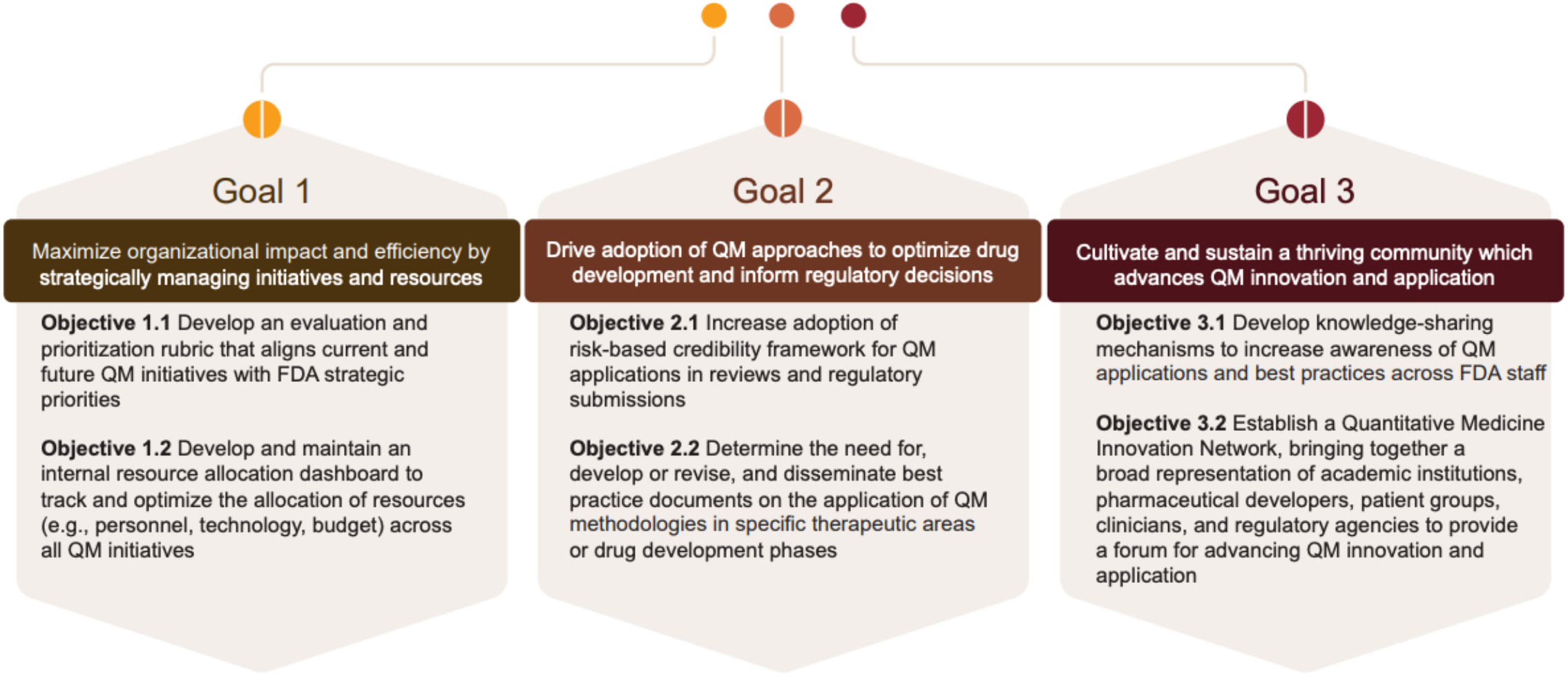
Provides a pathway for regulatory acceptance of dynamic tools for use in drug development programs

Disease Area	Submitter	Tool	Trial Component	Issuance Date and Supporting Information
Alzheimer's disease	The Coalition Against Major Diseases (CAMD)	Disease Model: Placebo/Disease Progression	Demographics, Drop-out	Issued June 12, 2013 • Determination Letter
Multiple	Janssen Pharmaceuticals and Novartis Pharmaceuticals	Statistical Method: MCP-Mod	Dose-Finding	Issued May 26, 2016 • Determination Letter • Statistical Review • Pharmacometric Review
Multiple	Ying Yuan, PhD The University of Texas MD Anderson Cancer Center Department of Biostatistics	Statistical Method: Bayesian Optimal Interval (BOIN) design	Dose-Finding	Issued: December 10, 2021 • Determination Letter • Statistical Review • Publication Erratum
Multiple	Pfizer	Statistical Method: Empirically Based Bayesian Emax Models	Dose-Finding	Issued: August 5, 2022 • Determination Letter • Multidisciplinary Review

R package: ClinDR

https://www.fda.gov/drugs/development-approval-process-drugs/drug-development-tools-fit-purpose-initiative

Scope of CDER QM CoE



[FDA/Critical Path Institute (C-Path) Model-Informed Drug Development Training Course]

[CDER QM CoE 2025 Progress and Priorities report]

MODEL-INFORMED DRUG DEVELOPMENT (MIDD) INTRODUCTION

What is MIDD (model-informed drug development)

University of Minnesota
Driven to Discover®

Describe the application of various quantitative models that leverage and integrate understanding of physiology, disease, and pharmacology to facilitate the decision-making process during drug development

- PK–PD models
- PBPK models
- QSP models
- Disease models
- Competitor information and meta-analysis
- Design and trial execution models
- Data analysis models
- Quantitative decision criteria
- Trial performance metrics

Other similar terms: model-based drug development (MBDD), model-informed drug discovery and development (MID31), and quantitative pharmacology (QP)

Steps to implement MIDD

Key contributors to MIDD

- MIDD plan is a clinical team deliverable, and Clinical Pharmacology, Statistics, and Pharmacometrics & Systems Pharmacology leads drive the completion of the plan with key input from clinical and other lines as appropriate

Key information to be included

- Key development questions and decisions that are informed by the analysis
- Contributing data
- Key assumptions
- Risks and limitations
- Impact to the development program focusing on the potential benefits (decreased risk, time, and increased efficiency)
- Potential impact if the analysis is not completed

Sahasrabudhe V, et al. Impact of Model-Informed Drug Development on Drug Development Cycle Times and Clinical Trial Cost. *Clin Pharmacol Ther*. 2025;118(2):378-385.

Benefits of implementing MIDD

- Sample size reduction
- Waivers of clinical trials
- "No-Go" decisions for conducting a trial
- Leading to cost and time savings

How much can MIDD save?

- Overall, the use of MIDD yielded annualized average savings of approximately 10 months of cycle time and $5 million per program
- Early Development: 5 months and $0.5 million per program
- Late Development: 22 months and $4.1 million per program
  - PopPK and ER analyses in pediatric programs produced the most substantial time savings (up to 4 years) by eliminating the need for dedicated Phase 1 PK/PD studies in pediatric populations

Sahasrabudhe V, et al. Impact of Model-Informed Drug Development on Drug Development Cycle Times and Clinical Trial Cost. *Clin Pharmacol Ther*. 2025;118(2):378-385.

How much does clinical trial cost?

University of Minnesota
Driven to Discover®

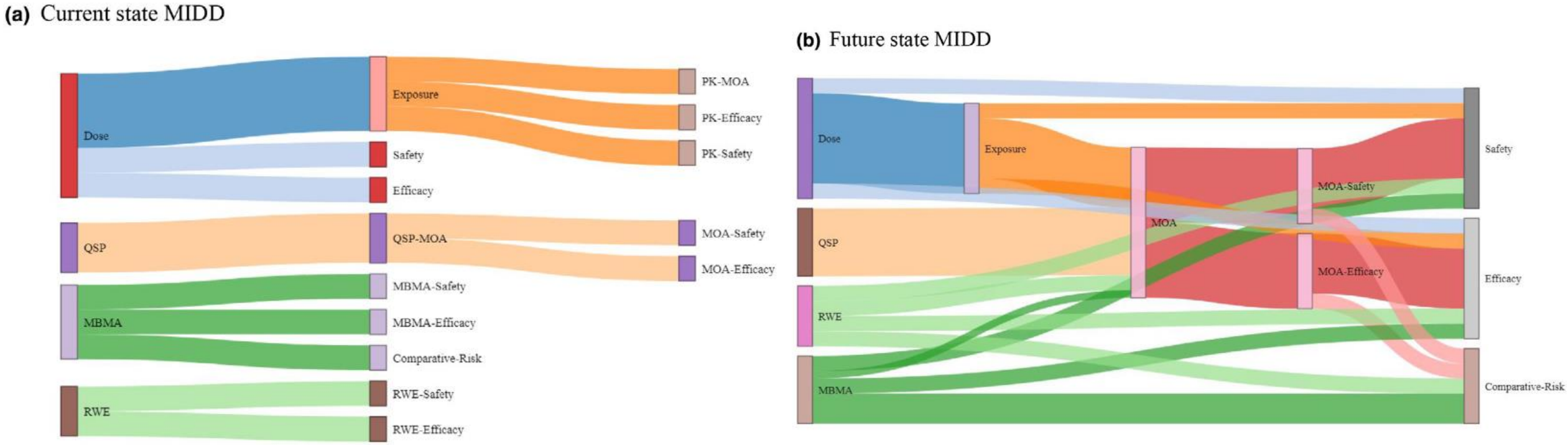
Table 1 Study timelines and average clinical trial budget estimates for typical Pfizer Phase I studies

Study type	Number of subjects, site	Protocol to CSR timeline (months)	Average clinical trial budget ($)
Bioavailability/Bioequivalence	24, PCRU	9	0.5 M
Thorough QT	36, PCRU	9	0.65 M
Renal impairment	24, external CRU	18	2 M
Hepatic impairment	24, external CRU	18	1.5 M
ADME	6, external CRU	9	1.5 M
Drug–drug interaction	12, PCRU	9	0.4 M

CRU, Clinical Research Unit; PCRU, Pfizer Clinical Research Unit.

Sahasrabudhe V, et al. Impact of Model-Informed Drug Development on Drug Development Cycle Times and Clinical Trial Cost. *Clin Pharmacol Ther*. 2025;118(2):378-385.

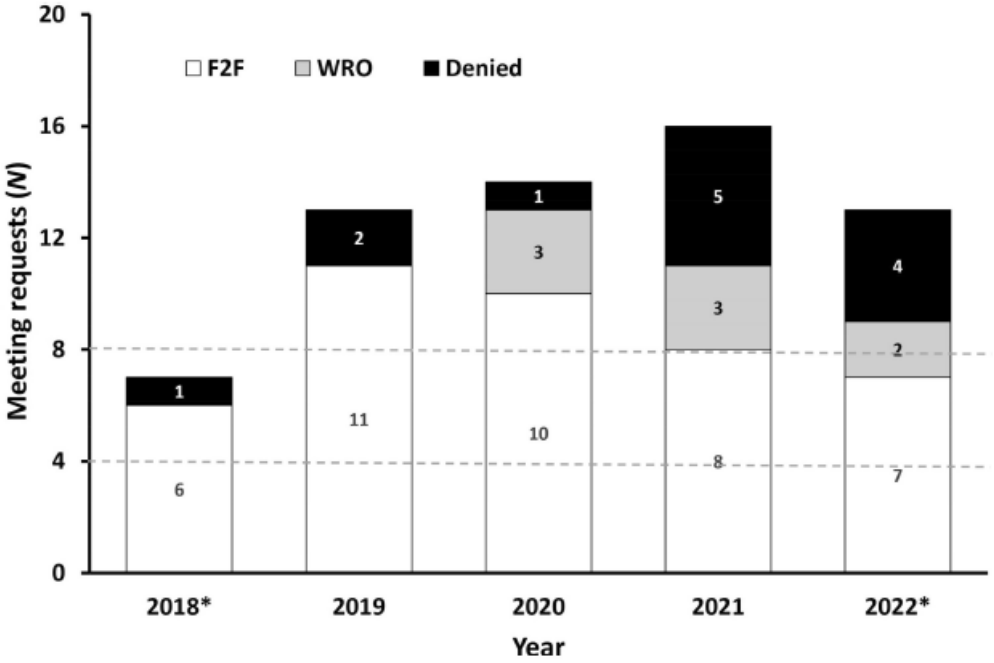
Future MIDD integrates different modalities into a holistic interface



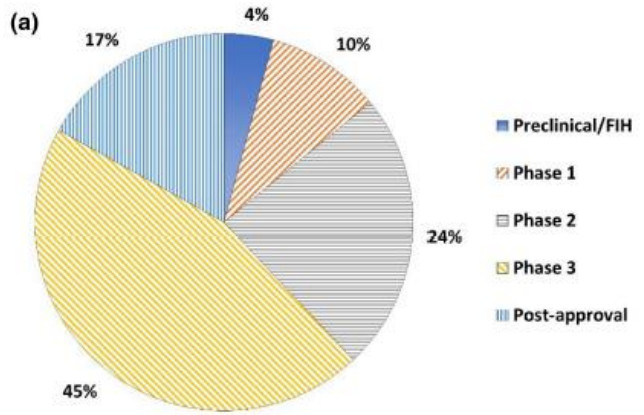
Clinical trial simulations would be enabled for each node or combination of nodes to interrogate future clinical questions (e.g., changing dosing regimen, different formulation characteristics, changes in patient characteristics)

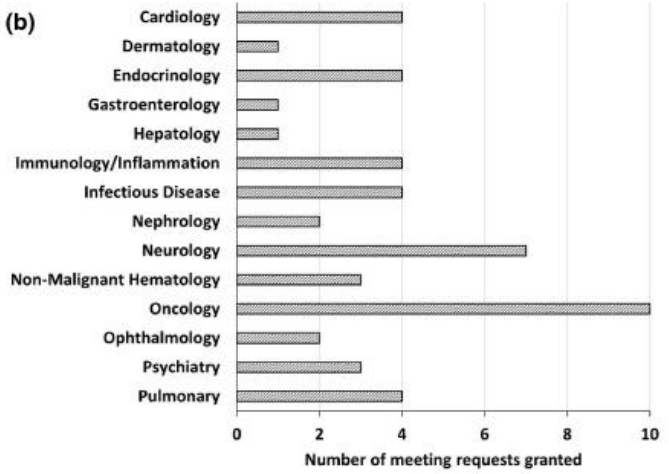
Sahasrabudhe V, et al. Impact of Model-Informed Drug Development on Drug Development Cycle Times and Clinical Trial Cost. *Clin Pharmacol Ther*. 2025;118(2):378-385.

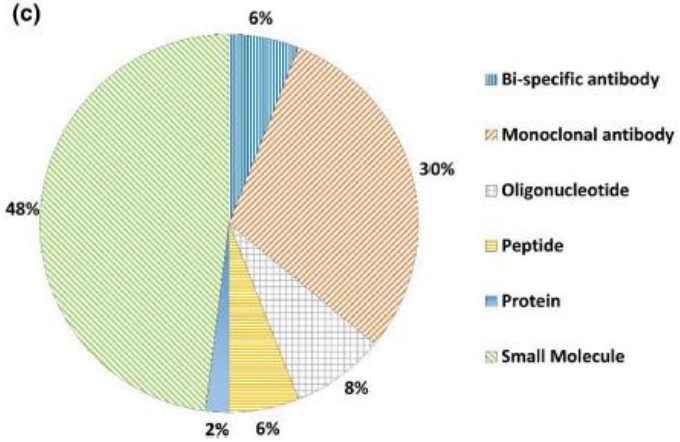
Increasing numbers of MIDD meeting requests with FDA

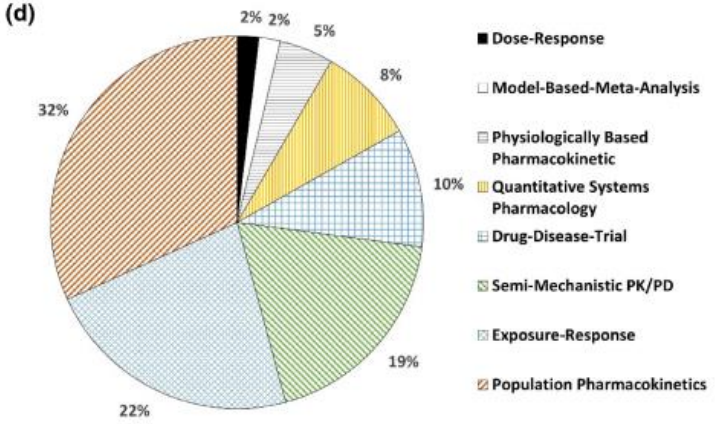


A total of 50 meeting requests (~ 10/year) between July 2018-Sep 2022







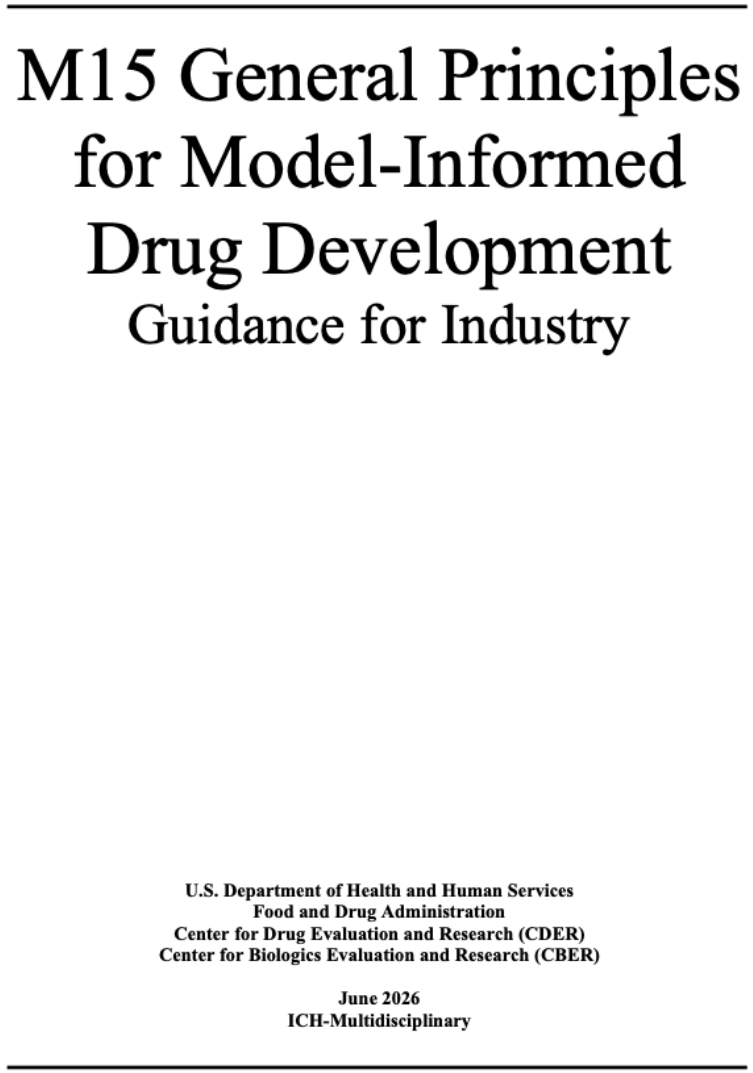


Madabushi R, et al. The US Food and Drug Administration's Model-Informed Drug Development Meeting Program: From Pilot to Pathway. *Clin Pharmacol Ther*. 2024;116(2):278-281.

BEST PRACTICES ON MIDD

Regulatory agencies have been promoting quantitative tools through publishing various guidance

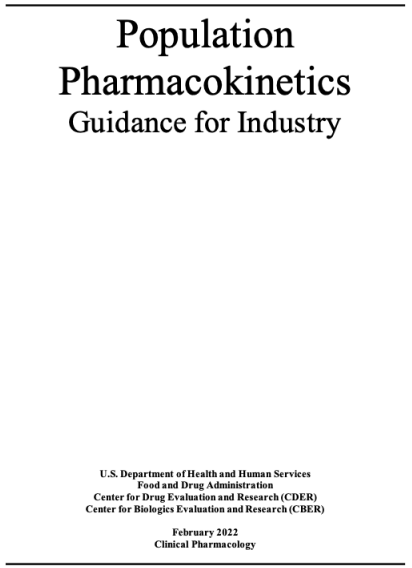
UNIVERSITY OF MINNESOTA
Driven to Discover®



M15 General Principles for Model-Informed Drug Development
Guidance for Industry

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)

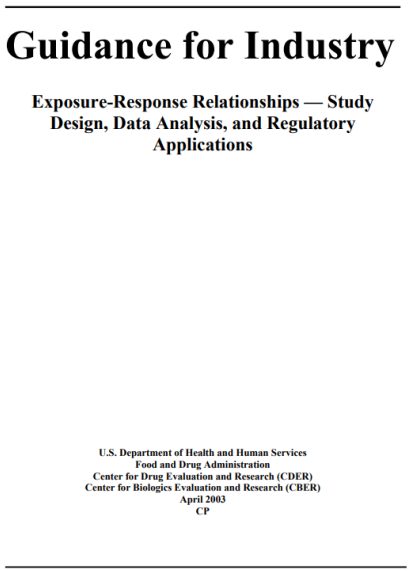
June 2026
ICH-Multidisciplinary



Population Pharmacokinetics
Guidance for Industry

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)

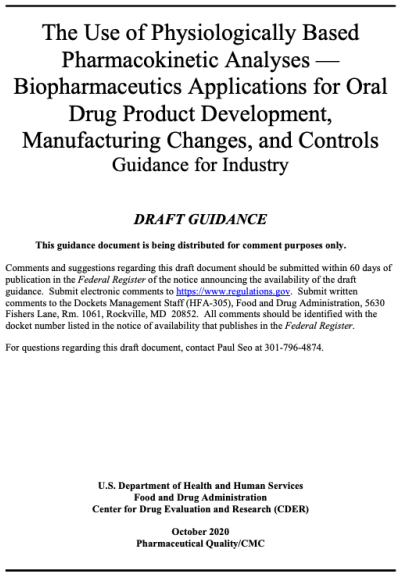
February 2022
Clinical Pharmacology



Guidance for Industry

Exposure-Response Relationships — Study Design, Data Analysis, and Regulatory Applications

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)
April 2003
CP



The Use of Physiologically Based Pharmacokinetic Analyses — Biopharmaceutics Applications for Oral Drug Product Development, Manufacturing Changes, and Controls
Guidance for Industry

DRAFT GUIDANCE

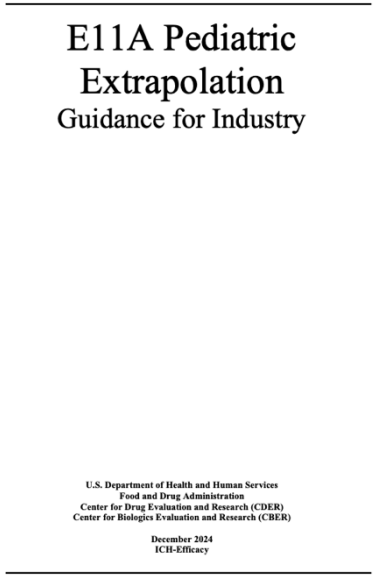
This guidance document is being distributed for comment purposes only.

Comments and suggestions regarding this draft document should be submitted within 60 days of publication in the *Federal Register* of the notice announcing the availability of the draft guidance. Submit electronic comments to https://www.regulations.gov. Submit written comments to the Dockets Management Staff (HFA-305), Food and Drug Administration, 5630 Fishers Lane, Rm. 1061, Rockville, MD 20852. All comments should be identified with the docket number listed in the notice of availability that publishes in the *Federal Register*.

For questions regarding this draft document, contact Paul Seo at 301-796-4874.

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)

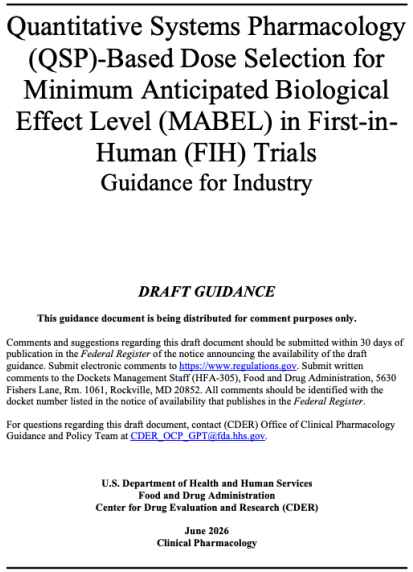
October 2020
Pharmaceutical Quality/CMC



E11A Pediatric Extrapolation
Guidance for Industry

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)

December 2024
ICH-Efficacy



Quantitative Systems Pharmacology (QSP)-Based Dose Selection for Minimum Anticipated Biological Effect Level (MABEL) in First-in-Human (FIH) Trials
Guidance for Industry

DRAFT GUIDANCE

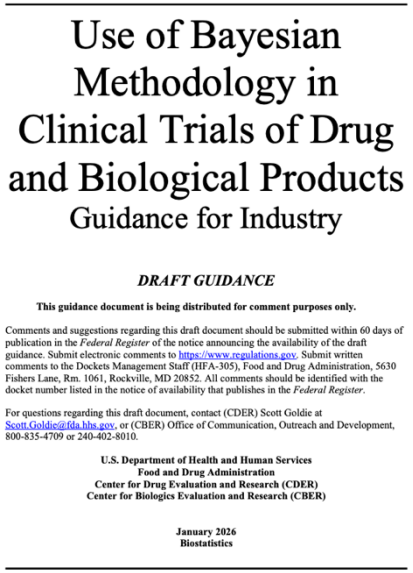
This guidance document is being distributed for comment purposes only.

Comments and suggestions regarding this draft document should be submitted within 30 days of publication in the *Federal Register* of the notice announcing the availability of the draft guidance. Submit electronic comments to https://www.regulations.gov. Submit written comments to the Dockets Management Staff (HFA-305), Food and Drug Administration, 5630 Fishers Lane, Rm. 1061, Rockville, MD 20852. All comments should be identified with the docket number listed in the notice of availability that publishes in the *Federal Register*.

For questions regarding this draft document, contact (CDER) Office of Clinical Pharmacology Guidance and Policy Team at CDER_OCP_GPT@fda.hhs.gov.

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)

June 2026
Clinical Pharmacology



Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products
Guidance for Industry

DRAFT GUIDANCE

This guidance document is being distributed for comment purposes only.

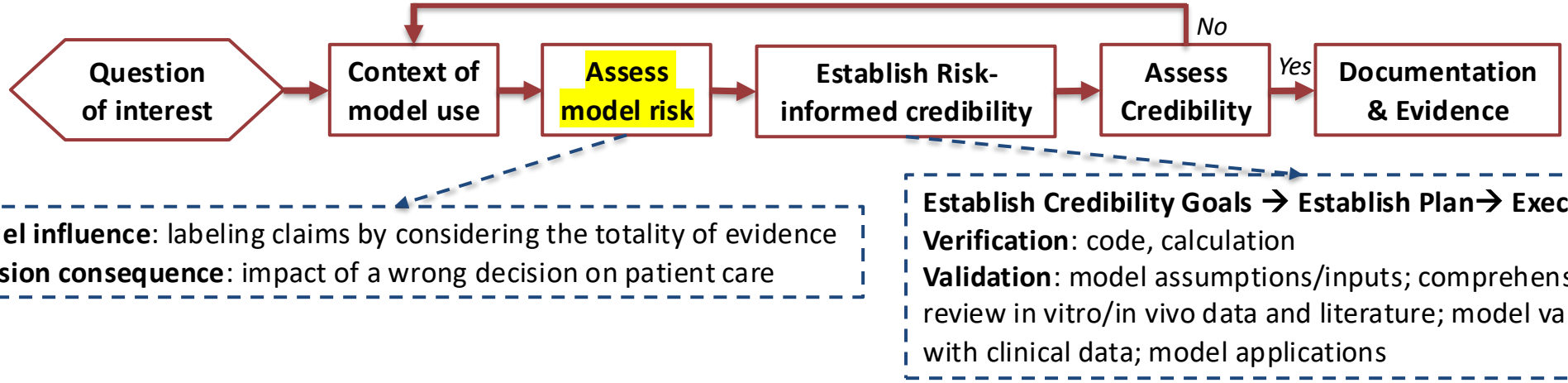
Comments and suggestions regarding this draft document should be submitted within 60 days of publication in the *Federal Register* of the notice announcing the availability of the draft guidance. Submit electronic comments to https://www.regulations.gov. Submit written comments to the Dockets Management Staff (HFA-305), Food and Drug Administration, 5630 Fishers Lane, Rm. 1061, Rockville, MD 20852. All comments should be identified with the docket number listed in the notice of availability that publishes in the *Federal Register*.

For questions regarding this draft document, contact (CDER) Scott Goldie at Scott.Goldie@fda.hhs.gov, or (CBER) Office of Communication, Outreach and Development, 800-835-4709 or 240-402-8010.

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)

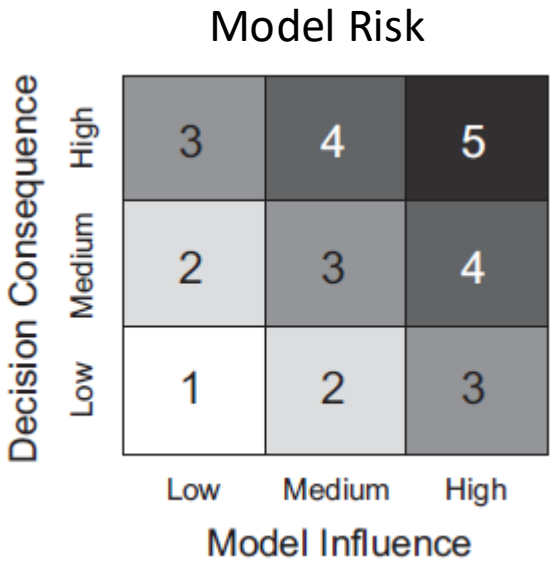
January 2026
Biostatistics

Credibility assessment framework in MIDD



Model influence	Description
Low	Model provides minor evidence; substantial nonclinical and clinical data are available to inform the decision
Medium	Model provides supportive evidence; some clinical trial data are available to inform the decision
High	Model provides substantial evidence; no clinical trial data relevant to the context of use or limited clinical trial data from similar scenarios are available to inform the decision

Decision consequence	Description
Low	Incorrect decision would not result in adverse outcomes in patient safety or efficacy
Medium	Incorrect decision could result in minor to moderate adverse outcomes in patient safety or efficacy
High	Incorrect decision could result in severe adverse outcomes in patient safety or efficacy



Kuemmel C, et al. Consideration of a Credibility Assessment Framework in Model-Informed Drug Development: Potential Application to Physiologically-Based Pharmacokinetic Modeling and Simulation. *CPT Pharmacometrics Syst Pharmacol*. 2020;9(1):21-28.

Assessment of MIDD Evidence

Item	Definition	Instruction	Entry
Key Assessment Elements			
Question of Interest[1]	The question that MIDD is intended to answer.	State the Question of Interest.	
Context of Use	A description of the model(s) and its specific role and scope to answer the Question of Interest.	Provide a concise, clear, and explicit description of the model, the data used to build the model, the specific role of the model outcomes, and the other data or evidence that will contribute to the answer to the Question of Interest.	
Model Influence	The intended weight of the model outcomes in decision-making considering the contribution of other relevant information.	Describe the Model Influence; rate it as low, medium, or high considering other relevant information (e.g., nonclinical and clinical) to inform decision-making; and justify the rating.	
Consequence of Wrong Decision	The consequences (e.g., with respect to patient safety and/or efficacy) if a wrong decision is made, based on all available information.	Describe the consequence of a wrong decision; rate it as low, medium, or high based on the severity of the consequences a wrong decision may have on patient safety and efficacy; and justify the rating.	
Model Risk[2]	The contribution of the model outcomes to a possible wrong decision and subsequent potential undesirable consequences.	Describe the risk; rate it as low, medium, or high based on the Model Influence rate and the Consequence of a Wrong Decision rate; and justify the rating.	
Model Impact	The contribution of the model outcomes in relation to current regulatory expectations or standards in answering the Question of Interest.	Describe the impact; rate it as low, medium, or high considering current regulatory expectations or standards; and justify the rating.	

[1] If MIDD is planned to answer different Questions of Interest, it is recommended to use separate tables for each question.

[2] Model Risk should be interpreted in the context of answering a specific Question of Interest and is not to be perceived as a risk intrinsic to MIDD or M&S.

Credibility assessment framework



Category	Context of use
Question of interest	How should the investigational drug be dosed when coadministered with CYP3A4 modulators?
Context of use	• Simulation to predict effects of weak and moderate CYP3A4 modulators on investigational drug PK • Predictions will be used for dosing recommendations in label • No DDI studies proposed with weak and moderate CYP3A4 modulators; have clinical data with strong CYP3A4 modulators
Model risk	**High**
Model influence	*High* • Model provides substantial evidence • Limited clinical data from similar scenarios to support the decision
Decision consequence	*Medium* • Incorrect decision could result in minor to moderate adverse patient outcomes
Validation plan	• Ensure model reproduces clinical PK data at different doses from healthy volunteers • Reproduce clinical PK data when dosed with strong CYP3A4 modulators • Reproduce effects observed for other CYP3A4 substrates with weak and moderate modulators

Kuemmel C, et al. Consideration of a Credibility Assessment Framework in Model-Informed Drug Development: Potential Application to Physiologically-Based Pharmacokinetic Modeling and Simulation. *CPT Pharmacometrics Syst Pharmacol*. 2020;9(1):21-28.

MIDD Planning Stage

MIDD Planning Stage[3] The following items/rows are to be completed at the MIDD Planning Stage.			
Appropriateness of Proposed MIDD	The rationale for why the proposed MIDD is suitable to answer the Question of Interest and cover the related key assumptions and required data.	Include a description and justification sufficient to facilitate regulatory interaction on the appropriateness of the proposed MIDD to answer the Question of Interest.	
Technical Criteria	A summary and rationale of the key criteria for Model Evaluation and model outcomes to establish the acceptability of the model (e.g., using an acceptance standard such as bioequivalence acceptance limits).	Include a description of the Technical Criteria for the assessment of Model Evaluation and model outcome. This should include sufficient details on the relevant metric(s).	

[3] These items should also be provided at the MIDD Evidence Submission Stage.

MIDD Evidence Submission Stage

Item	Definition	Instruction	Entry
MIDD Evidence Submission Stage The following items/rows are to be filled at the MIDD Evidence Submission Stage after data collection and execution of the model.			
Model Evaluation	A brief discussion of the key results and conclusions of the technical evaluation[4] of the model.	Describe the key results and how they compare to and fulfill the Technical Criteria and conclude on the acceptability of the model performance and model outcome, with details being provided in the appropriate regulatory documentation (see Section 4).	
Outcome of the MIDD Evidence Assessment[5]	A concise summary of the multidisciplinary assessment of the MIDD evidence to answer the Question of Interest.	Provide a multidisciplinary integrative assessment and conclusion for the acceptability of the MIDD evidence to contribute to the answer to the Question of Interest, referring to the MIDD assessment framework elements.	

[4] Using the principles of Model Evaluation described in Section 3, with specific focus on Technical Criteria.
[5] "Assessment" in this context does not refer to any regulatory review activities or processes.

Model Analysis Report (MAR) Content

Sections	Content
Executive Summary	• An overview of the rationale for the analyses • A brief summary of the data and methods • A brief summary of the results and conclusions
Introduction	• The rationale for the analyses • Relevant background information and knowledge • If applicable, a description of pre-existing analyses with reference to previously submitted reports
Objectives	The objectives of the analyses
Data and Methods	Descriptions of the following: • Data sources ○ Criteria and rationale with respect to source data inclusion and exclusion ○ Relevant design features of studies and/or experiments • M&S methods, model development, and strategic approaches (e.g., the sequence of development, numerical methods, and Technical Criteria; see Section 2 and Section 3) • Approaches for Model Evaluation (i.e., verification, validation, and applicability; see Section 3) • If applicable, prediction and simulation methods and scenarios

Model Analysis Report (MAR) Content

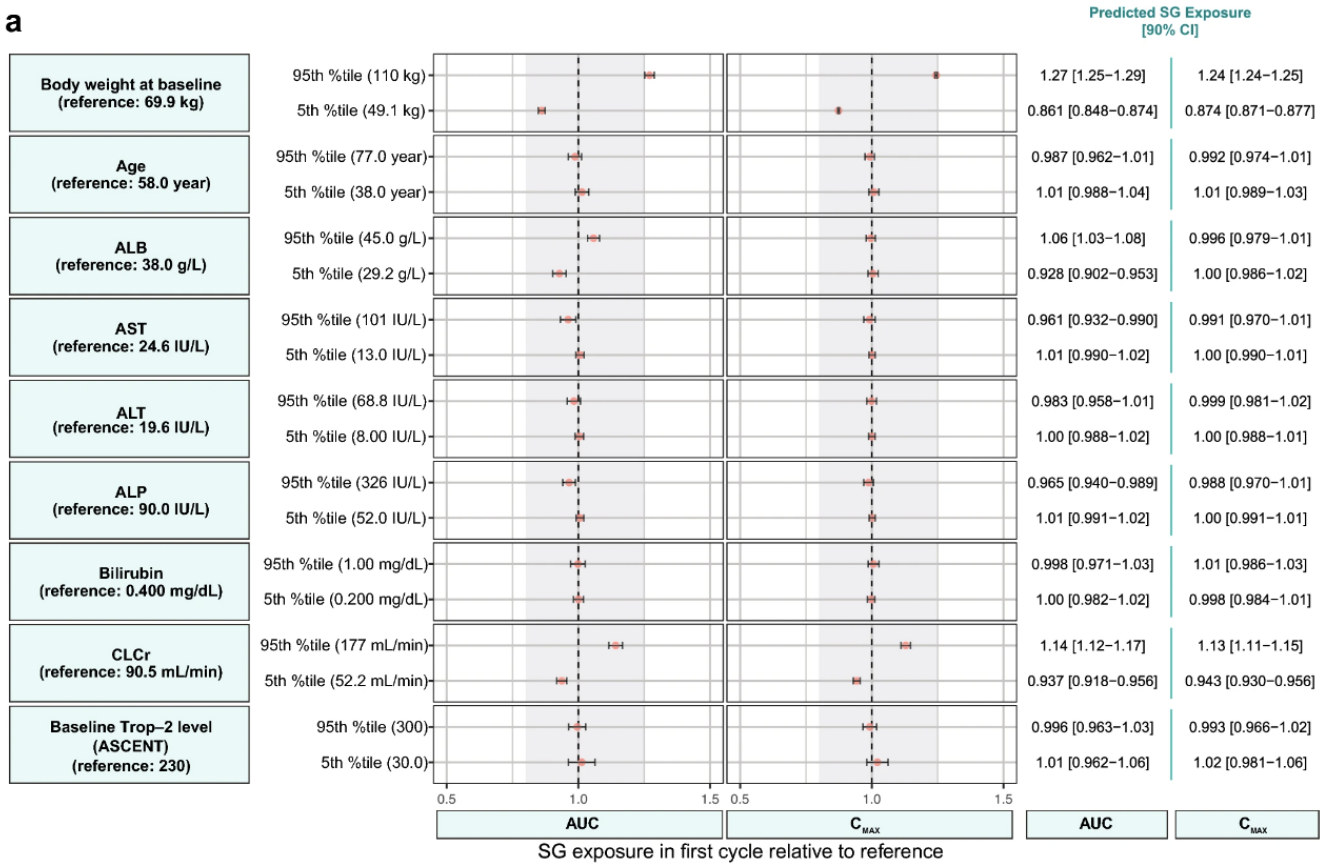
Results	• Data description, including graphical and/or tabular displays, as appropriate. Data excluded during the analyses should be described along with appropriate rationale. • The results, including graphical and/or tabular displays, of model development and Model Evaluation, with predictions and simulations, if applicable • Any deviations from the MAP should be described and justified.
Discussion	Interpretation of results, including the adequacy, potential limitations of the data and M&S, and clinical and/or other implications, taking into account: • Deviations from the MAP • Model Evaluation (including Technical Criteria and applicability of the model) • Relevant nonclinical and clinical information and knowledge, if applicable
Conclusions	The conclusions of the analyses
References	A references list covering the sources of data used for the analyses (e.g., bioanalytical reports, clinical study reports, laboratory reports, or literature)
Appendices	Additional materials cross-referenced in the MAR, for example: • Supplemental data descriptions and model development and evaluation results, including graphical and/or tabular displays, as appropriate • The user-generated code for the key model(s)

EXAMPLES

Interpret covariate effects using PopPK modeling

Question: Sacituzumab govitecan (Trodelvy) is approved for locally advanced and metastatic breast cancer. Does sacituzumab govitecan dose need to be adjusted for intrinsic or extrinsic factors?

Context of use: Use PopPK to examine the impact of covariates.



Hepatotoxicity according to NCI common toxicity criteria

Grade	1	2	3	4
AST, ALT (IU/L)	≤2.5xULN	>2.5-5xULN	>5-20xULN	>20xULN
Total bilirubin (mg/dL)		<1.5xULN	>1.5-3xULN	>3xULN

USPI: *No adjustment to the starting dosage is required when administering TRODELVY to patients with mild hepatic impairment [see Clinical Pharmacology (12.3)].*

Sathe AG, et al. Population Pharmacokinetics of Sacituzumab Govitecan in Patients with Metastatic Triple-Negative Breast Cancer and Other Solid Tumors. *Clin Pharmacokinet.* 2024;63(5):669-681.

Pediatric approval based on exposure-matching between pediatric and adult with PopPK modeling

University of Minnesota
Driven to Discover®

Question: Ceftaroline is a broad-spectrum cephalosporin with in vitro activity against Gram-positive bacteria associated with cSSTI, including *S. aureus,* streptococci, and β-lactamase–producing Gram-negative species. High dose ceftaroline fosamil (600 mg Q8H, adjusted for eCrCL ≤50 mL/min) was approved in adults with cSSTI caused by *S. aureus* with MIC of 2-4 mg/L. What is the dose recommendation for pediatric patients (aged ≥2 months to < 18 years)?

Context of use: Use PopPK to match predicted pediatric exposures to the adult population and compare with observed data in other pediatric indications.

Age group	Ceftaroline fosamil dose (2-h i.v. infusions every 8 h)	$C_{max,ss}$ (mg/L)[a]	$AUC_{24,ss}$ (mg/L*h)[a]	$C_{max,ss}$ ratio to adults	$AUC_{24,ss}$ ratio to adults	35% *f*T>MIC of 2 mg/L	35% *f*T>MIC of 4 mg/L
Normal renal function							
Adults	600 mg	18.4 (10.4, 32.2)	155 (85.7, 285.0)	–	–	99.7	82.7
12 to <18 years	12 mg/kg (max 600 mg)	21.7 (12.6, 35.9)	173 (99.1, 299.0)	1.18	1.12	99.8	90.2
6 to <12 years		23.5 (14.5, 37.5)	178 (106, 302.0)	1.28	1.15	99.8	91.8
2 to <6 years		21.4 (13.2, 33.9)	153 (90.9, 258.0)	1.16	0.987	99.5	81.8
12 to <24 months	10 mg/kg	19.2 (11.9, 30.4)	146 (86.9, 247.0)	1.04	0.940	99.7	80.8
2 to <12 months		20.3 (12.6, 32.0)	168 (98.9, 284.0)	1.11	1.08	99.9	90.8

%*f*T>MIC, percent of time that free plasma concentration is above minimum inhibitory concentration

[a]Median (5th, 95th percentiles) based on a summary of 100 trials; corresponds to median (90% prediction interval)

Impact: selected appropriate high dose regimens in pediatric patients to shorten study duration

cSSTI: complicated skin and soft-tissue infections

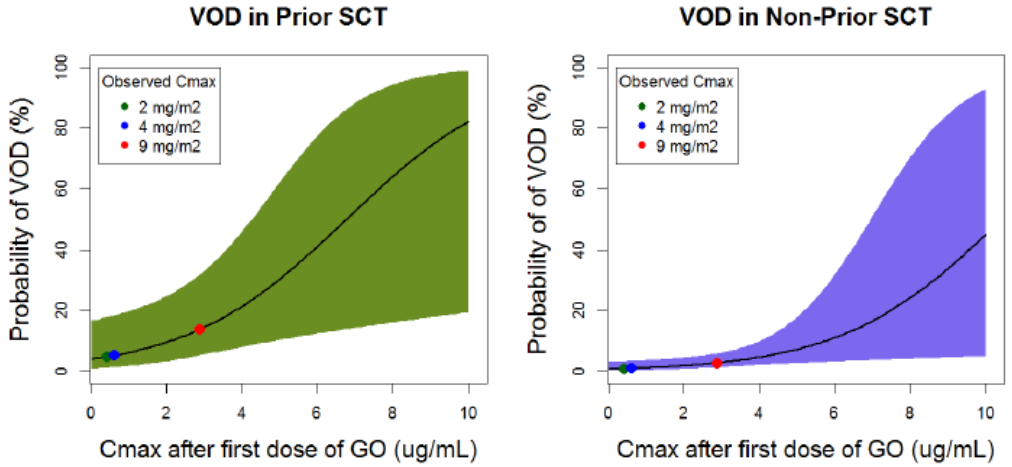
Chan PLS, et al. The use of extrapolation based on modeling and simulation to support high-dose regimens of ceftaroline fosamil in pediatric patients with complicated skin and soft-tissue infections. *CPT Pharmacometrics Syst Pharmacol*. 2021 Jun;10(6):551-563.

Support a lower fractionated dose of Mylotarg using ER analysis

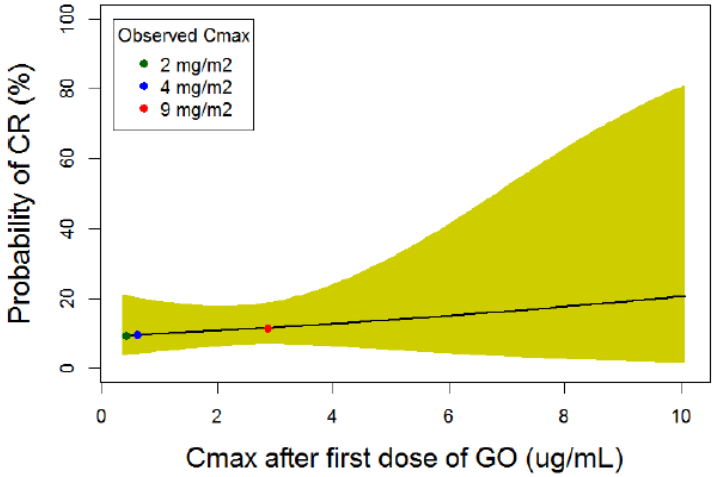
Question: Gemtuzumab ozogamicin (Mylotarg) received accelerated approval in 2000, but a Phase 3 study (SWOGS106) failed to confirm benefit with higher rate of hepatic veno-occlusive disease (VOD) which has substantial morbidity and mortality. Does changing induction dose from 9 mg/m^2 on Day 1/15 to 3 mg/m^2 on Day 1/4/7 maintain the same efficacy but reduce safety?

Context of use: Use ER analyses to support lower fractionated dose.

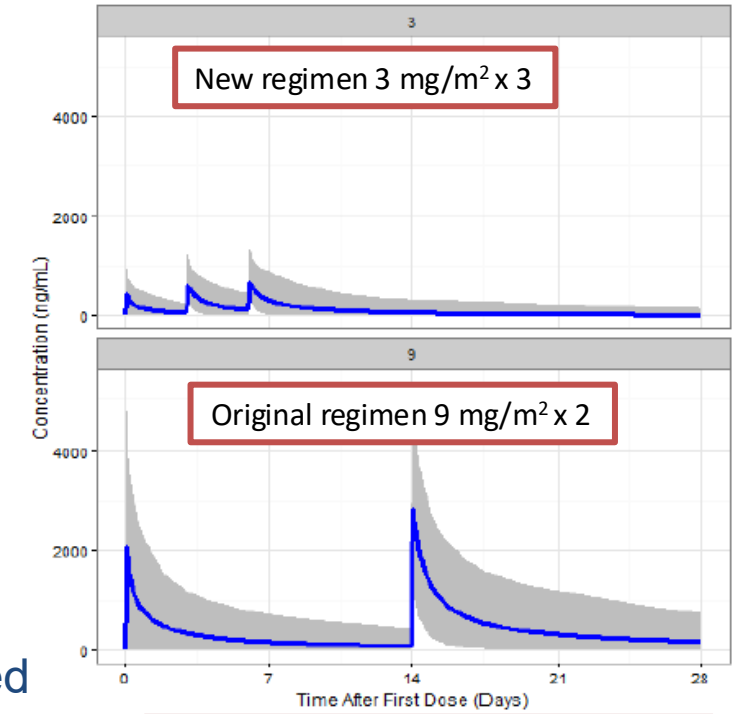
Significant ER safety



Flat ER efficacy



CD33 site saturation achieved at Dose > 2 mg/m^2



Cmax reduced by ~75%

FDA Clinical Pharmacology and Biopharmaceutics Reviews. 761060Orig1s000 and 761060Orig2s000

Predicting survival using tumor growth model

Question: Tumor size-based end points, such objective response rates (ORRs) or even progression-free survival (PFS), often demonstrated a weak correlation with overall survival (OS). Is there a better surrogate endpoints that can better forecast OS and to quantify drug effect in patients with cancer?

Context of use: Use the tumor growth inhibition (TGI) – OS model to predict OS and then to bridge the gap between early and late phase drug development.

TGI model

Wang's Model

$$TS_i(t) = TS_i(0) \bullet e^{-SR_i \bullet t} + PR_i \bullet t$$

TS, tumor size (sum of the longest diameters)
SR, tumor shrinkage rate
PR, tumor progression rate

Chan's Model

$$TS_i(t) = TS_i(0) \bullet \left(e^{-KS_i \bullet t} + e^{KG_i \bullet t} - 1\right)$$

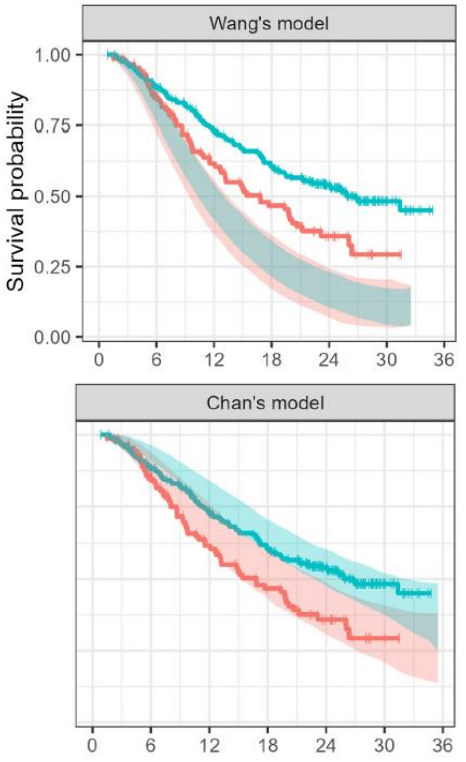
KS, tumor shrinkage rate
KG, tumor growth rate

Survival model

Cox regression model with:

- TGI metrics
- Baseline characteristics
  - Wang's model: ECOG, baseline tumor size, % tumor reduction from baseline at wk 8
  - Chan's model: log (KG), CRP, # metastatic sties, race, Alb, LDH, IC or TC > 0 vs. IC and TC = 0, Neutrophil-to-Lymphocyte ratio, lines of therapy, liver metastasis, time-to-tumor growth

OS prediction

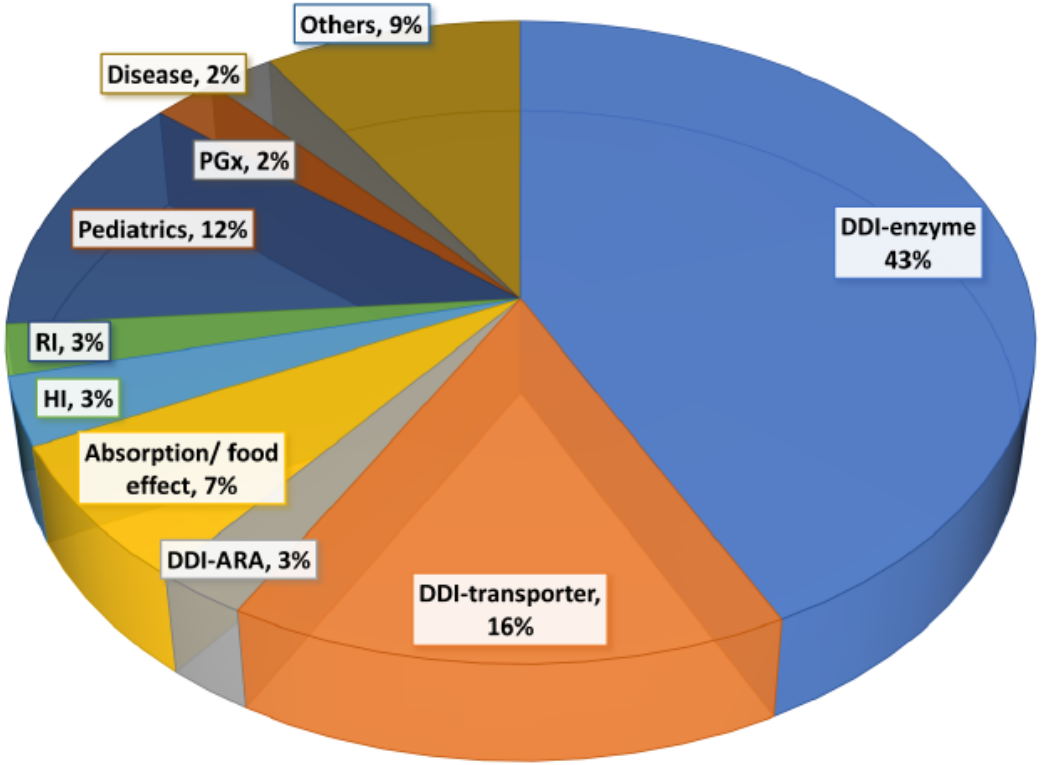


Sheng Y, et al. Tumor growth inhibition-overall survival modeling in non-small cell lung cancer: A case study from GEMSTONE-302. *CPT Pharmacometrics Syst Pharmacol*. 2024;13(3):437-448.

Increased use of PBPK to support NDA/BLA

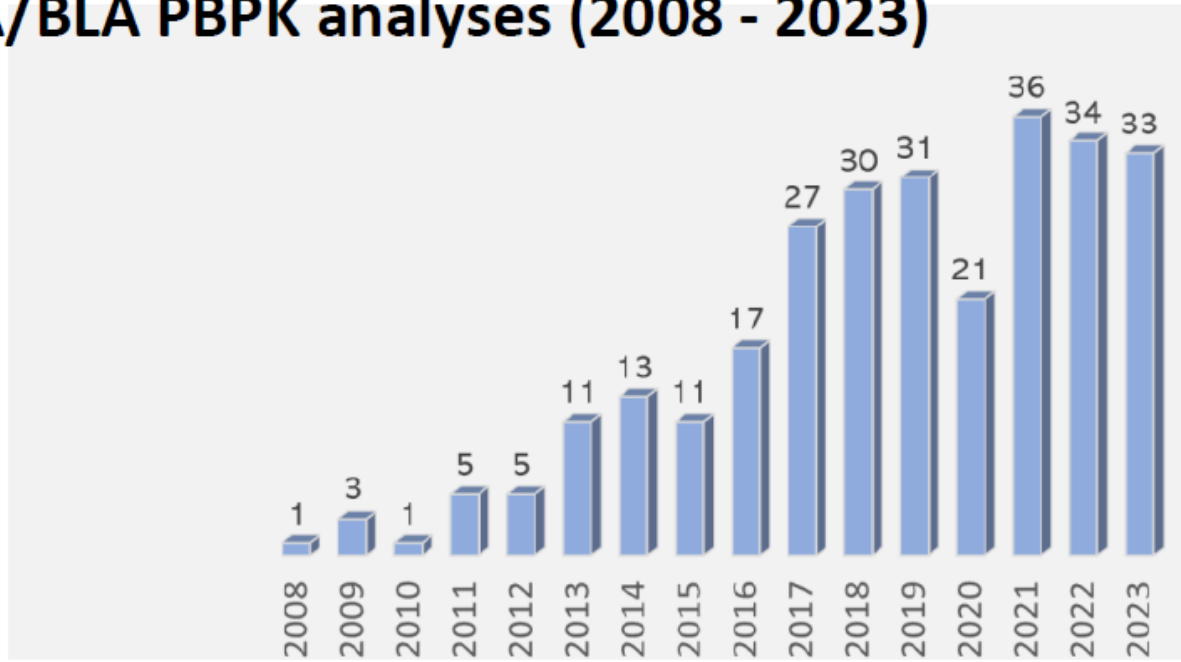
University of Minnesota
Driven to Discover®

Areas of Application in IND/NDA/BLA N=368, NDA=137 (2018- 2022)



- DDI applications constitute 60%
- Many analyses were submitted as INDs

NDA/BLA PBPK analyses (2008 - 2023)



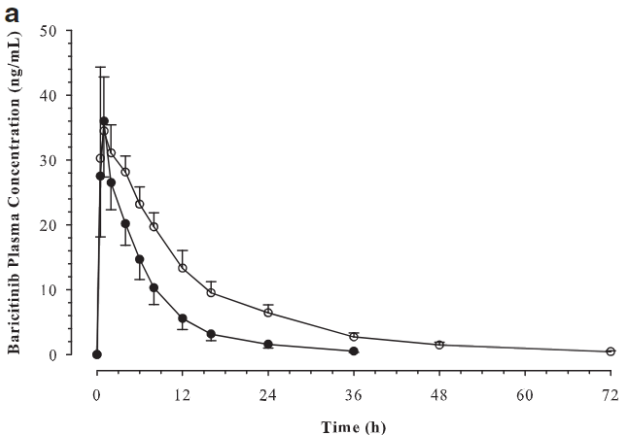
Previously presented in 'Current practice in model evaluation and assessment for PBPK models' reusability in new drug development ' May 3, 2024. MCERSI/FDA MMF meeting

Slide credit: Dr. Yuching Yang, ACCP 2024

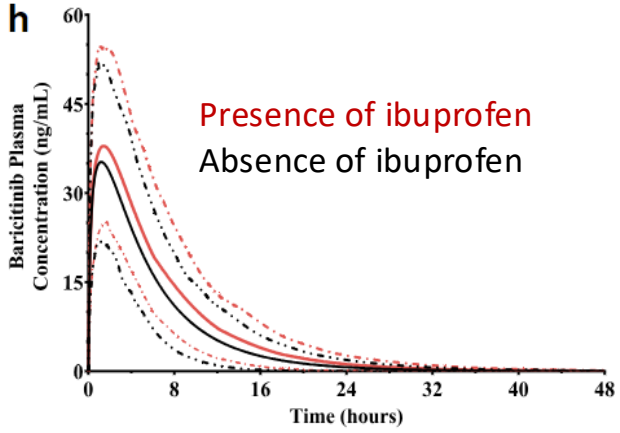
Predicting OAT3 mediated DDI using PBPK

Question: Baricitinib is a JAK inhibitor to treat adults with severe alopecia areata and with moderately to severely active rheumatoid arthritis after treatment with 1 or more TNF inhibitors. Baricitinib is a substrate for OAT3 MATE2-K, P-gp, and BCRP. What is the effect of ibuprofen (OAT3 inhibitor) on baricitinib PK?

Context of use: Use PBPK models based on probenecid (a strong OAT3 inhibitor) and baricitinib to predict the effect of ibuprofen on baricitinib.



Impact of probenecid on baricitinib (geo ratio and 90% CI):
Cmax: 1.03 (0.94, 1.13)
AUC0-inf: 2.03 (1.91, 2.16)



PBPK prediction (geo ratio and 90% CI):
Cmax: 1.17 (1.15–1.7)
AUCo-inf: 1.95 (1.84–2.07)

Table 4 Observed and predicted AUC and C_{max} for baricitinib, probenecid, ibuprofen and diclofenac.

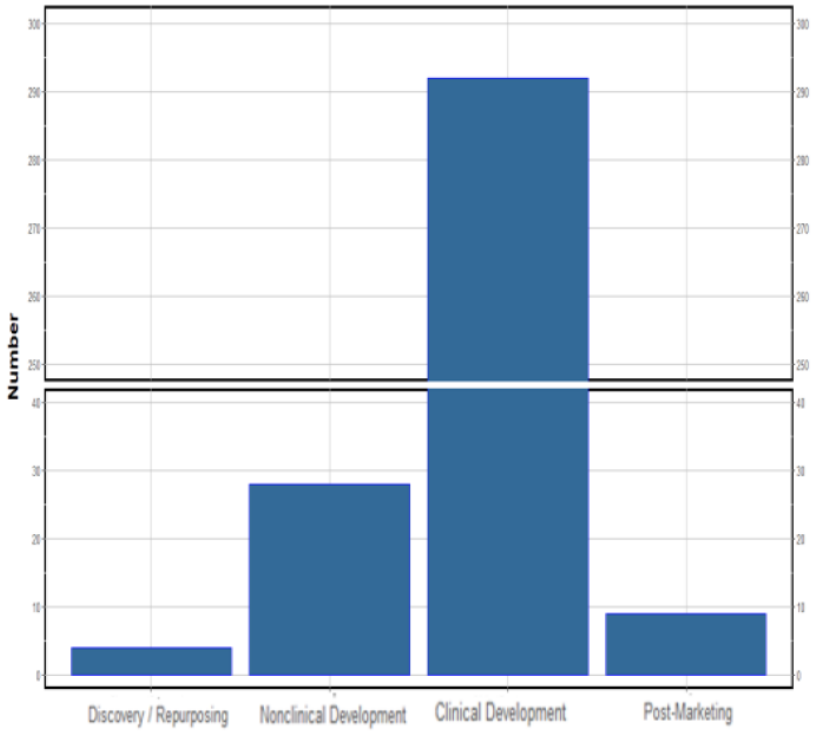
Compound	Parameter	Observed Mean (CV)	Predicted Geometric Mean (5-95% CI)	Observed/ Predicted
Baricitinib	C_{max} (ng/mL)	36.2 (22)	34.9 (33.3 – 36.7)	1.0
	AUC(0-∞) (ng·h/mL)	236 (22)	230 (215 – 247)	1.0
Probenecid (1,000 mg BID)	C_{max} (μg/mL)	145	160 (150 – 169)	0.9
	$C_{ss,ave}$ (μg/mL)	115	116 (94 – 139)	1.0
Ibuprofen (800 mg QID)	C_{max} (μg/mL)	55.6 (24)[a]	52.8 (50.1 – 55.6)	1.1
	AUC (μg·h /mL)	218 (23)[a]	183 (170 – 196)	1.2
Diclofenac(100 mg BID)	C_{max} (μg/mL)	1.2 – 3.6[b]	2.7	–
	AUC (μg·h/mL)	2.7 – 4.4[b]	3.0	–
Baricitinib (+/- Probenecid)	AUC ratio(90% CI)[c]	2.03(1.91-2.16)	1.95 (1.84-2.07)	1.0
Baricitinib (+/- Probenecid))	C_{max} ratio(90% CI)[d]	1.03 (0.940, 1.13)	1.17 (1.15-1.7)	0.9
Baricitinib (+/- Ibuprofen)	AUC ratio(90% CI)[c]	NA	1.24 (1.22-1.26)	NA
Baricitinib (+/- Ibuprofen)	C_{max} ratio(90% CI)[d]	NA	1.07 (1.06-1.08)	NA
Baricitinib (+/-Diclofenac)	AUC ratio (90% CI)[c]	NA	1.00 (1.00-1.00)	NA
Baricitinib (+/- Diclofenac)	C_{max} ratio(90% CI)[d]	NA	1.00 (1.00-1.00)	NA

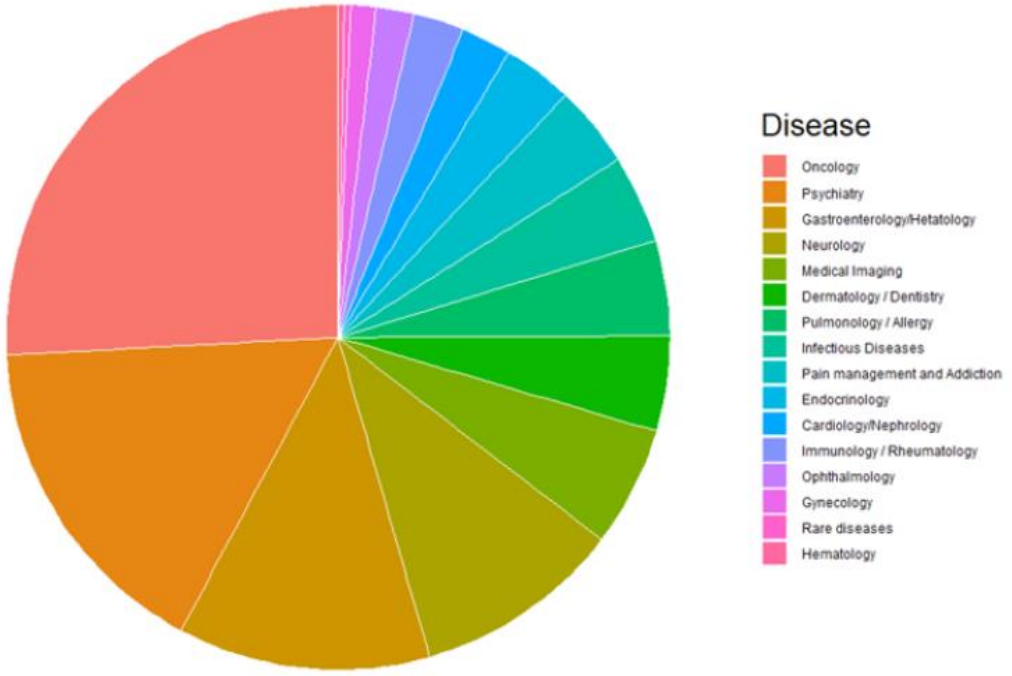
$AUC_{(0-\infty)}$ = area under the concentration-time curve (AUC) from zero to infinity; CI = confidence interval; C_{max} = maximum observed concentration; $C_{ss,ave}$ = average concentration at steady state.
[a]Published mean[45]
[b]Published ranges[17]
[c]AUC ratio = AUC of baricitinib in the presence of the inhibitor (probenecid, ibuprofen or diclofenac) / AUC of baricitinib alone
[d]Cmax ratio = Cmax of baricitinib in the presence of the inhibitor (probenecid, ibuprofen or diclofenac) / Cmax of baricitinib alone

USPI: *"…ibuprofen (OAT3 inhibitors with less inhibition potential) predicted minimal effect on the PK of baricitinb"*

Posada MM, et al. Prediction of Transporter-Mediated Drug-Drug Interactions for Baricitinib. *Clin Transl Sci*. 2017;10(6):509-519.

AI/ML submissions by development stage (2016-2022)





Regulatory submissions to the CDER that included AI/ML by stage of the drug development life cycle and therapeutic area in which it was proposed or applied, US FDA from 2016 to 2021.

Common analysis types

- Outcome prediction
- Covariate selection/confounding adjustment
- Pharmacometric modeling
- Dose selection/optimization
- Anomaly detection
- Imaging, video, and voice analysis
- RWD, NLP
- Drug discovery/repurposing
- Drug toxicity prediction
- Enrichment design
- Pt risk stratification
- Adherence
- Synthetic control
- Endpoint/biomarker assessment
- Postmarking surveillance

Liu Q, et al. Landscape Analysis of the Application of Artificial Intelligence and Machine Learning in Regulatory Submissions for Drug Development From 2016 to 2021. *Clin Pharmacol Ther*. 2023;113(4):771-774.

ML-based Patient Population Selection

Question: Patients in the Anakinra Phase 3 trial (SAVE-MORE) were selected based on a biomarker cut-off, soluble urokinase plasminogen activator receptor (suPAR) ≥ 6 ng/mL, intending to enrich the trial with patients at risk for progression to severe respiratory failure. SuPAR test is not commercially available in the United States at the time.

Context of use: Use ML to explore clinical characteristics and/or laboratory tests that could reliably identify the patients with baseline suPAR ≥ 6 ng/mL.

Table 1 Eight criteria in the scoring rule to identify patients with suPAR levels of 6 ng/mL or higher

Age ≥ 75 years
Severe pneumonia by WHO criteria
Current/previous smoking status
Sequential Organ Failure Assessment (SOFA) score ≥ 3
Neutrophil-to-lymphocyte ratio (NLR) ≥ 7
Hemoglobin ≤ 10.5 g/dL
Medical history of ischemic stroke
Blood urea ≥ 50 mg/dL and/or medical history of renal disease

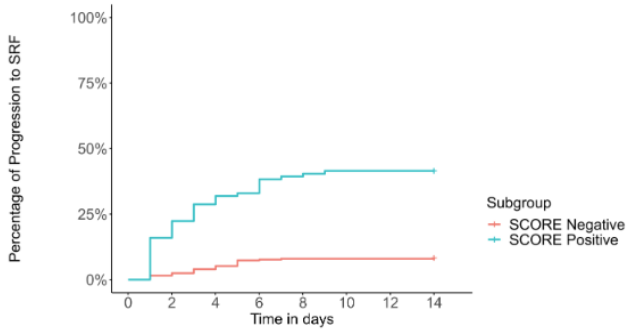
suPAR, soluble urokinase plasminogen activator receptor; WHO, World Health Organization.

The elastic net regression was used to select contributing feature(s) and a neural network-based model was applied to independently select feature(s) and the related cutoff value(s).

Table 2 Performance of the scoring rule to identify patients with suPAR levels of 6 ng/mL or higher in both the training dataset (SAVEMORE trial) and the external validation dataset (SAVE trial)

Training dataset (SAVEMORE trial)				External validation dataset (SAVE trial)			
	suPAR ≥ 6	suPAR < 6	Total		suPAR ≥ 6	suPAR < 6	Total
SCORE (+)	231	12	243	SCORE (+)	95	6	101
	(PPV = 0.95, sensitivity = 0.41)	(FPR = 0.04)			(PPV = 0.94, sensitivity = 0.37)	(FPR = 0.07)	
SCORE (–)	338	256	594	SCORE (–)	159	76	235
		(NPV = 0.43, specificity = 0.96)				(NPV = 0.32, specificity = 0.93)	
Total	569	268	837	Total	254	82	336

FPR, false positive rate; NPV, negative predictive value; PPV, positive predictive value; suPAR, soluble urokinase plasminogen activator receptor.



Clinical scoring rule had high positive predictive value, high specificity, reasonable sensitivity, and biological relevance. It is expected to identify patients who are likely to have an elevated suPAR level ≥ 6 ng/mL at baseline.

Liu Q, et al. Using Machine Learning to Determine a Suitable Patient Population for Anakinra for the Treatment of COVID-19 Under the Emergency Use Authorization. *Clin Pharmacol Ther*. 2024;115(4):890-895.

Concepts of model-based quantitative decision making

ECP 8506 Clinical Trial Simulation
Sep 18, 2026

Ya-Feng (Jay) Wen, PharmD, PhD
jay.wen3@gilead.com | wenxx164@umn.edu
Associate Clinical Pharmacology Director, Gilead Sciences